

Min cost max flow

Last update : July 11, 2026

Description:

Min cost max flow is the extend problem of max flow,addtional constraint is require least cost,and the purpose i write this is it got some interesting nature

Algorithm:

The algorithm is simple we run a **spfa** on residue graph and using the ways similar to **Edmond-karp algorithm**.

The reason we use spfa is since forward edge consists of positive weight but the reverse edge consists of negative weight so we need spfa,and there is a important nature here

Lemma 1:

For a residue graph,if we always choose the min cost path then it won't form negative cycle

We will prove by contradiction,assume that current residue graph G contain no negative cycle and we know let the min cost path from source S to sink T be P_{st} and G^R be the graph which remove P_{st} and all vertex on it and also edge connected to any vertex on it

By assumption we know G^R and P_{st} contain no negative cycle,then in next residue graph G^R remain unchange and all reverse edge on P_{st} will form so obviously if contain a negative cycle it must form by these reverse edge and G^R Assume thoes reverse edge of P_{st} be

$$T \rightarrow \cdots \rightarrow v \rightarrow u \rightarrow \cdots \rightarrow S$$

then we observe that if a vertex on reverse P_{st} go G^R and come back it either direct go back itself or another vertex before it on reverse P_{st} ,so WLOG assume that a path from u to G^R and come back on v and form a cycle,we can observe that this cycle contain two part,one part located at reverse P_{st} and another part located in G^R ,we define these two part as d_{G^R} and d_{vu} ,since they form a negative cycle then

$$d_{G^R} + d_{vu} < 0$$

but we know that $d_{G^R} \geq d_{uv}$,otherwise we have another much minimum path from u to v ,we can conclude that

$$\begin{aligned}
d_{uv} &= -d_{vu} \\
d_{G^R} &\geq d_{uv} \\
d_{G^R} &\geq -d_{vu} \\
d_{G^R} + d_{uv} &\geq 0
\end{aligned}$$

which is contradiction to our assumption thus it won't form negative cycle